Software Requirements Specification — Chat Application (Socket Programming)

Oct 12, 2026 · @Amogh

Document version 1.0 (draft). Prepared for the Software Engineering Mini-Project, Deliverables Part-1. Format follows IEEE Std 830-1998, with the security, use case and traceability sections required by the course.

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Chat Application (Socket Programming), a real-time text chat system. It states what the system shall do and the quality levels it shall meet.

The intended readers are the project team (developers, the Test Plan team, the Architecture and Design team) and the course faculty who evaluate the project.

1.2 Scope

The product is a web-based chat application in which users exchange text messages instantly over a persistent socket connection between a browser client and a server.

In scope

- User registration, login and logout.
- Creating, listing, joining and leaving chat rooms.
- Sending text messages and receiving them in real time, with sender name and timestamp.
- Showing which users are online in a room.
- Handling connection loss, reconnection and input errors.
- Storing messages so that recent history is shown when a user joins a room.

Out of scope for this version

- File, image, audio or video sharing.
- Voice or video calls.
- End-to-end encryption.
- Native mobile applications and push notifications.
- Administrator and moderation features.

Two features are listed as optional (Section 3.7): a typing indicator and private one-to-one messaging. They are not part of the committed scope.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
TC	Test Case
RTM	Requirements Traceability Matrix
Client	The web page running in the user's browser
Server	The backend program that authenticates users and relays messages
Socket	An endpoint of a two-way network connection between client and server
WebSocket	A protocol (RFC 6455) that provides a persistent, full-duplex connection over a single TCP connection
Socket.IO	A library that provides event-based real-time communication, using WebSocket as its main transport
Chat room (room)	A named group conversation; a message sent to a room is delivered to its current members
Guest	A person who has not logged in
User	A person who has registered and logged in
Session token	A value issued by the server at login that identifies an authenticated user
HTTP / HTTPS	Hypertext Transfer Protocol / HTTP over TLS
WS / WSS	WebSocket / WebSocket over TLS
TLS	Transport Layer Security

Term	Meaning
XSS	Cross-Site Scripting
JSON	JavaScript Object Notation
LAN	Local Area Network
Proposed target	A numerical value chosen by the team because the project brief does not specify one

1.4 References

1. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
2. ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.
3. I. Fette and A. Melnikov, RFC 6455: The WebSocket Protocol, IETF, 2011.
4. Socket.IO documentation, version 4.
5. OWASP, Password Storage Cheat Sheet.
6. *Software Engineering Mini-Project Deliverables Part-1* (course handout).
7. Mini-project topic list, entry 5: Chat Application (Socket Programming).

1.5 Overview of the Document

Section	Content
2	Overall description: context, functions, users, environment, constraints, assumptions
3	Functional requirements (FR-01 to FR-23)
4	Non-functional requirements (NFR-01 to NFR-10)
5	External interface requirements, including the client-server socket protocol
6	UML use case diagram and use case descriptions
7	Security objectives and security requirements
8	Requirements Traceability Matrix
9	Requirements quality check
Appendix A	Consolidated list of all requirement IDs

Section	Content
Appendix B	Compliance checklist against the assignment

2. Overall Description

2.1 Product Perspective

The Chat Application is a new, self-contained system. It does not replace or extend an existing product and does not depend on any third-party chat service.

It follows a client-server model with three parts:

Part	Role
Client	A web page in the user's browser. It shows the user interface and exchanges events with the server.
Server	A single server process. It authenticates users, manages rooms and connections, and relays each message to the members of the room.
Database	Stores user accounts, rooms and messages.

The client uses HTTP for registration and login, and a persistent WebSocket-based connection for all real-time chat events. Clients never communicate directly with each other; every message passes through the server.

2.2 Product Functions

Function group	Summary
Account management	Register, log in, log out
Room management	Create a room, list rooms, join a room, leave a room
Messaging	Send a text message; receive messages from other users in real time; view recent history
Presence	See which users are online in the current room; see join and leave notices
Connection handling	Detect connection loss, show the connection state, reconnect automatically

Function group	Summary
Error handling	Reject invalid input and show a clear error message

2.3 User Classes and Characteristics

User class	Description	Technical skill expected	Main functions used
Guest	A person who has opened the application but has not logged in	Basic web browsing	Register, log in
User	A registered person who has logged in	Basic web browsing; familiar with any messaging app	All room, messaging and presence functions; log out

There is no separate administrator class in this version. The development team maintains the server directly.

2.4 Operating Environment

Element	Environment
Client	A current stable release of Google Chrome, Mozilla Firefox or Microsoft Edge with JavaScript enabled
Client operating system	Any operating system that runs one of the browsers above (tested on Windows 10 and 11)
Server	Node.js (current LTS release) on Windows or Linux (assumption A-01)
Database	SQLite, stored as a file on the server machine (assumption A-02)
Network	TCP/IP; a LAN or localhost for development and demonstration

2.5 Design and Implementation Constraints

ID	Constraint
C-01	Real-time communication shall use socket/WebSocket-based communication, as fixed by the project title. Periodic HTTP polling written by the team is not acceptable as the main mechanism.
C-02	The technology stack shall be one of the two options in the topic list: JavaScript (Node.js with WebSockets) or Python (Flask-SocketIO).
C-03	The system shall be built with free and open-source tools only.
C-04	The system shall run on a single server machine; no cloud infrastructure or load balancer is required.
C-05	The scope shall be achievable by a student team within one semester.
C-06	Documentation shall follow the IEEE formats named in the course deliverables.

2.6 Assumptions and Dependencies

The items below are assumptions made by the SRS author, not confirmed project decisions. If one changes, the affected requirements must be reviewed.

ID	Assumption or dependency	Affects
A-01	The team selects Node.js with the Socket.IO library. If Flask-SocketIO is chosen instead, the requirements remain valid; only Sections 2.4 and 5.2 change.	Sections 2.4, 5.2
A-02	SQLite is used for storage. Any relational or document database that persists users, rooms and messages is acceptable.	Section 5.2, FR-13
A-03	User authentication (registration and login) is implemented. The project brief does not state this explicitly.	FR-01 to FR-04, Section 7

ID	Assumption or dependency	Affects
A-04	Development, testing and demonstration take place on localhost or a college LAN, with at most 20 concurrent users.	NFR-01, NFR-05, NFR-07
A-05	All numerical values in Sections 3, 4 and 7 are proposed targets set by the team, since the brief specifies none.	All measurable requirements
A-06	Users have a working network connection to the server and a supported browser.	Section 2.4
A-07	Only text messages are exchanged.	FR-09
D-01	The system depends on the Node.js runtime, the Socket.IO library, a password-hashing library (bcrypt) and the SQLite driver.	Section 5.2

3. System Requirements: Functional Requirements

Each requirement has a unique ID and a priority. **Must** requirements form the committed scope. **Should** requirements are planned but may be dropped if time runs short. **Optional** requirements are proposals outside the committed scope.

All numerical limits are proposed targets (assumption A-05).

3.1 Account Management

ID	Requirement	Priority
FR-01	The system shall allow a Guest to register an account by submitting a username of 3 to 20 characters (letters, digits and underscore only) and a password of 8 to 64 characters.	Must
FR-02	The system shall reject a registration whose username already exists, compared without regard to letter case, and shall display the message "Username already taken".	Must
FR-03	The system shall authenticate a Guest who submits a registered username and its correct password, issue a session token, and display the room list.	Must

ID	Requirement	Priority
FR-04	The system shall allow a User to log out. On logout the system shall invalidate the session token, close the user's socket connection and display the login page.	Must

3.2 Room Management

ID	Requirement	Priority
FR-05	The system shall allow a User to create a chat room with a name of 3 to 30 characters, and shall reject a name that already exists with the message "Room name already exists".	Must
FR-06	The system shall display to a User the list of all existing chat rooms.	Must
FR-07	The system shall allow a User to join one chat room selected from the room list. A connection shall be a member of at most one room at a time.	Must
FR-08	The system shall allow a User to leave the current chat room. After leaving, the User shall receive no further messages from that room.	Must

3.3 Messaging

ID	Requirement	Priority
FR-09	The system shall allow a User to send a text message of 1 to 500 characters to the room the User has joined.	Must
FR-10	The server shall deliver each accepted message to every User currently connected to that room, including the sender, without a page reload.	Must
FR-11	The system shall allow two or more Users to be members of the same room at the same time, and shall display the sender's username with every message.	Must

ID	Requirement	Priority
FR-12	The server shall assign a timestamp to each message when it is received, and the client shall display that timestamp with the message in the viewer's local time in HH:MM format.	Must
FR-13	The system shall store every accepted message, and when a User joins a room shall display the 50 most recent messages of that room in chronological order.	Should

3.4 Presence

ID	Requirement	Priority
FR-14	The system shall display the usernames of all Users currently online in the joined room, and shall update this list whenever a User joins, leaves or disconnects.	Must
FR-15	The system shall display a notice in the room, visible to the other members, when a User joins or leaves it, in the form "<username> joined the room" or "<username> left the room".	Should

3.5 Connection and Disconnection Handling

ID	Requirement	Priority
FR-16	The client shall display the current connection state as one of "Connected", "Reconnecting" or "Disconnected", and shall disable the send control while the state is not "Connected".	Must
FR-17	When the connection is lost, the client shall attempt to reconnect automatically at least 5 times. On success it shall rejoin the room the User was in. If all attempts fail it shall display "Connection lost. Please reload the page."	Must
FR-18	The server shall remove a User from the online list of the room within 60 seconds of that user's connection being lost.	Must

3.6 Error Handling

ID	Requirement	Priority
FR-19	The system shall reject a message that is empty, contains only whitespace, or exceeds 500 characters. A rejected message shall not be delivered to any User, and an error message shall be displayed to the sender only.	Must
FR-20	When login fails, the system shall display the message "Invalid username or password" and shall not indicate which of the two was wrong.	Must
FR-21	When a request fails because of a server-side error, the client shall display an error message that contains no stack trace or internal detail, and shall remain usable without a page reload.	Should

3.7 Optional Features (not in committed scope)

ID	Requirement	Priority
FR-22	The system shall display "<username> is typing" to the other members of a room while a User is entering a message, and shall remove it within 3 seconds after typing stops.	Optional
FR-23	The system shall allow a User to send a private text message to one other online User, visible only to the sender and that recipient.	Optional

4. Non-Functional Requirements

The project brief specifies no numerical quality targets. Every value below is a **proposed target** chosen by the team for a LAN or localhost demonstration (assumptions A-04 and A-05). The "Verification" column states how each requirement is measured.

ID	Category	Requirement	Verification
NFR-01	Performance	With 20 concurrent Users on a LAN, the system shall display at least 95% of messages on every recipient's screen within 1 second of the sender submitting them.	Send 100 messages; compare send and receive times logged by the clients.
NFR-02	Performance	The system shall complete a login request and display the room list within 2 seconds on a LAN.	Time 10 logins with the browser developer tools.
NFR-03	Reliability	While sender and recipients stay connected, the system shall deliver 100% of accepted messages, in the order the server received them, with no duplicates.	Send 100 numbered messages; check each recipient shows all 100 once, in order.

ID	Category	Requirement	Verification
NFR-04	Reliability	When the network is restored within 30 seconds of a connection loss, the client shall re-establish the connection within 10 seconds of restoration without any user action.	Disable and re-enable the client's network; time the state indicator.
NFR-05	Availability	The server shall run continuously for a 2-hour session with 20 concurrent Users without crashing or requiring a restart.	2-hour soak test with simulated clients; inspect the server log.
NFR-06	Usability	At least 4 of 5 first-time users shall be able to register, join a room and send a message within 3 minutes, without instructions.	Observed trial with 5 students outside the team.
NFR-07	Scalability	A single server instance shall support at least 20 concurrent connected Users across at least 5 rooms while still meeting NFR-01.	Load test with 20 simulated clients spread over 5 rooms.
NFR-08	Maintainability	The source code shall be organised into separate modules for client interface, authentication, socket event handling and data access, and a team member shall be able to install and start the system in under 15 minutes by following the README.	Code review against the module list; timed installation on a clean machine.
NFR-09	Security	The system shall satisfy every Must-priority security requirement in Section 7.2.	Security test cases in the Test Plan, Section 5.1.

ID	Category	Requirement	Verification
NFR-10	Compatibility	Every Must-priority functional requirement shall work on the current stable releases of Google Chrome, Mozilla Firefox and Microsoft Edge on Windows 10 or 11, at a window width of 1024 pixels or more.	Run the functional test cases once on each browser.

5. External Interface Requirements

5.1 User Interface

The user interface is a set of web pages. Layout and visual style are left to the design team; the elements below are required.

Screen	Required elements	Related requirements
Registration	Username field, password field, Register button, link to Login, error message area	FR-01, FR-02
Login	Username field, password field, Login button, link to Registration, error message area	FR-03, FR-20
Room list	List of rooms with a Join control for each, room name field with Create button, Logout button	FR-04 to FR-07
Chat room	Room name, message area showing sender, text and timestamp for each message, online user list, message input field, Send button, Leave button, connection state indicator, error message area	FR-08 to FR-19

General interface rules:

- All interface text shall be in English.
- Pressing Enter in the message input field shall have the same effect as the Send button.
- When a new message arrives, the message area shall scroll to show it.
- Every error message shall be shown as text near the control that caused it.

5.2 Software Interfaces

Names and versions depend on assumptions A-01 and A-02 and will be confirmed in the Software Architecture and Design Specification.

Interface	Software	Purpose
Server runtime	Node.js, current LTS release	Runs the server program
Real-time library	Socket.IO, version 4, on server and client	Event-based messaging over a WebSocket connection
Database	SQLite	Stores users (username, password hash), rooms (name) and messages (room, sender, text, timestamp)
Password hashing	bcrypt library	Hashes and verifies passwords (SEC-REQ-01)
Client platform	Web browser with JavaScript and WebSocket support	Runs the client

5.3 Communication Interfaces

The client and server communicate over TCP/IP in two ways.

HTTP for account operations. Registration, login and logout are ordinary request-response exchanges with JSON bodies. A successful login returns a session token.

A socket connection for chat. After login, the client opens one persistent connection to the server using Socket.IO, which uses the WebSocket protocol (RFC 6455) as its transport. The steps are:

1. The client sends an HTTP request asking to upgrade the connection to WebSocket, and includes the session token.
2. The server validates the token. If it is invalid, the server refuses the connection (SEC-REQ-02).
3. If the token is valid, the connection stays open. Either side can now send data at any time, without a new request.
4. Client and server exchange named events with JSON payloads, listed in the table below.
5. The server relays each accepted message to the connections that belong to the same room.

6. The library exchanges periodic heartbeat packets. If they stop, each side treats the connection as lost (FR-16, FR-18).
7. On logout, tab close or network failure the connection is closed, and the server updates the online list.

Note: Socket.IO can fall back to HTTP long-polling when a WebSocket connection cannot be established. This is library behaviour and satisfies constraint C-01.

Proposed socket events. Event names are proposals; the design team may rename them, provided the Test Plan is updated.

Event	Direction	Payload	Purpose	Requirement
`join_room`	Client to server	roomName	Join a room	FR-07
`leave_room`	Client to server	none	Leave the current room	FR-08
`send_message`	Client to server	text	Send a message to the current room	FR-09
`new_message`	Server to client	sender, text, timestamp	Deliver a message to room members	FR-10 to FR-12
`room_history`	Server to client	list of up to 50 messages	Show recent messages on joining	FR-13
`user_list`	Server to client	list of usernames	Update the online user list	FR-14, FR-18
`user_joined`, `user_left`	Server to client	username	Show a join or leave notice	FR-15
`error_message`	Server to client	code, message	Report a rejected request to the sender	FR-19, FR-21

The sender's identity is never part of a client payload. The server takes it from the authenticated connection (SEC-REQ-03).

Proposed HTTP endpoints.

Method and path	Purpose	Requirement
POST /api/register	Create an account	FR-01, FR-02

Method and path	Purpose	Requirement
POST /api/login	Authenticate and return a session token	FR-03, FR-20
POST /api/logout	Invalidate the session token	FR-04
GET /api/rooms	Return the room list	FR-06
POST /api/rooms	Create a room	FR-05

Ports and encryption. The server listens on one configurable TCP port (default 3000, a proposal) for both HTTP and socket traffic. On localhost the system uses HTTP and WS. If it is deployed on a shared network, HTTPS and WSS are to be used (SEC-REQ-08).

5.4 Hardware Interfaces

The system has no direct hardware interface. It needs only a standard computer with a network adapter for the server, and a keyboard, pointing device and display for each client. All hardware access is handled by the operating system and the browser.

6. UML Use Case Diagram

6.1 Actors

Actor	Type	Description
Guest	Primary	A person who has opened the application but has not logged in
User	Primary	A registered person who has logged in

A Guest becomes a User by completing UC-02.

6.2 Use Cases

ID	Use case	Actor	Functional requirements
UC-01	Register	Guest	FR-01, FR-02
UC-02	Log In	Guest	FR-03, FR-20
UC-03	Create Chat Room	User	FR-05
UC-04	Join Chat Room	User	FR-06, FR-07, FR-13, FR-15

ID	Use case	Actor	Functional requirements
UC-05	Send Message	User	FR-09, FR-19
UC-06	Receive Messages	User	FR-10, FR-11, FR-12
UC-07	View Online Users	User	FR-14, FR-18
UC-08	Leave Chat Room	User	FR-08, FR-15
UC-09	Log Out	User	FR-04
UC-10	Reconnect After Connection Loss	User	FR-16, FR-17

6.3 Relationships

Relationship	Type	Meaning
Guest to UC-01, UC-02	Association	A Guest can register and log in
User to UC-03 through UC-10	Association	A User can perform every chat function
UC-04 to UC-07	Include	Joining a room always displays its online users
UC-10 to UC-06	Extend	Reconnection extends receiving messages, only when the connection is lost

6.4 Diagram (Mermaid source)

Mermaid has no dedicated use case diagram type, so the diagram is written as a flowchart: circles are actors, rounded shapes are use cases, the box is the system boundary, solid lines are associations and dashed arrows are include and extend relationships.

```
flowchart LR
    Guest((Guest))
    User((User))

    subgraph SYS[Chat Application]
        UC01([UC-01 Register])
        UC02([UC-02 Log In])
        UC03([UC-03 Create Chat Room])
        UC04([UC-04 Join Chat Room])
        UC05([UC-05 Send Message])
        UC06([UC-06 Receive Messages])
```

```
        UC07([UC-07 View Online Users])
        UC08([UC-08 Leave Chat Room])
        UC09([UC-09 Log Out])
        UC10([UC-10 Reconnect After Connection Loss])
    end

    Guest --- UC01
    Guest --- UC02

    UC03 --- User
    UC04 --- User
    UC05 --- User
    UC06 --- User
    UC07 --- User
    UC08 --- User
    UC09 --- User
    UC10 --- User

    UC04 -. "include" .-> UC07
    UC10 -. "extend" .-> UC06
```

6.5 Use Case Descriptions

UC-01 Register

Field	Description
Use Case ID	UC-01
Use Case Name	Register
Actor	Guest
Preconditions	The Guest has opened the Registration page. The server is running.
Main Flow	(1) The Guest enters a username and a password. (2) The Guest selects Register. (3) The system validates the format of both fields. (4) The system checks that the username is not in use. (5) The system stores the username and the hashed password. (6) The system displays the Login page with a success message.

Field	Description
Alternative / Exception Flow	3a. A field has an invalid format: the system shows which rule failed and creates no account. 4a. The username exists: the system shows "Username already taken" and creates no account.
Postconditions	A new account exists. The Guest is not yet logged in.

UC-02 Log In

Field	Description
Use Case ID	UC-02
Use Case Name	Log In
Actor	Guest
Preconditions	The Guest has a registered account and has opened the Login page.
Main Flow	(1) The Guest enters username and password. (2) The Guest selects Login. (3) The system verifies the credentials. (4) The system issues a session token. (5) The client opens the socket connection using the token. (6) The system displays the room list.
Alternative / Exception Flow	3a. The credentials are wrong: the system shows "Invalid username or password" and stays on the Login page. 3b. Five consecutive failures for this username: the system blocks further attempts for 5 minutes (SEC-REQ-07). 5a. The socket connection fails: the system shows the state "Disconnected" and UC-10 begins.
Postconditions	The Guest is now a User with a valid session token and an open socket connection.

UC-03 Create Chat Room

Field	Description
Use Case ID	UC-03

Field	Description
Use Case Name	Create Chat Room
Actor	User
Preconditions	The User is logged in and is on the Room list page.
Main Flow	(1) The User enters a room name. (2) The User selects Create. (3) The system validates the name and checks that it is unique. (4) The system stores the room. (5) The system shows the new room in the room list.
Alternative / Exception Flow	3a. The name is shorter than 3 or longer than 30 characters: the system shows an error and creates no room. 3b. The name exists: the system shows "Room name already exists".
Postconditions	The new room exists and can be joined by any User.

UC-04 Join Chat Room

Field	Description
Use Case ID	UC-04
Use Case Name	Join Chat Room
Actor	User
Preconditions	The User is logged in, has an open socket connection, and at least one room exists.
Main Flow	(1) The User selects Join on a room in the room list. (2) The client sends a join request over the socket. (3) The server adds the connection to the room. (4) The system displays the Chat room page with the 50 most recent messages. (5) The system displays the online user list (UC-07). (6) The other members see a join notice.
Alternative / Exception Flow	2a. The room no longer exists: the system shows an error and stays on the room list. 4a. The room has no messages: the message area is empty.
Postconditions	The User is a member of the room and appears in its online user list.

UC-05 Send Message

Field	Description
Use Case ID	UC-05
Use Case Name	Send Message
Actor	User
Preconditions	The User has joined a room and the connection state is "Connected".
Main Flow	(1) The User types a message. (2) The User selects Send or presses Enter. (3) The client sends the text over the socket. (4) The server validates the text. (5) The server adds the sender's username and a timestamp, and stores the message. (6) The server delivers the message to all members of the room (UC-06). (7) The client clears the input field.

Field	Description
Alternative / Exception Flow	4a. The text is empty, whitespace only or over 500 characters: the server rejects it, the sender sees an error, and nothing is delivered. 3a. The connection is lost before sending: the send control is disabled and UC-10 begins.
Postconditions	The message is stored and visible to all members of the room.

UC-06 Receive Messages

Field	Description
Use Case ID	UC-06
Use Case Name	Receive Messages
Actor	User
Preconditions	The User has joined a room and the connection state is "Connected".
Main Flow	(1) Another member sends a message (UC-05). (2) The server pushes the message over the User's socket connection. (3) The client displays the sender's username, the text and the timestamp at the end of the message area, without a page reload.
Alternative / Exception Flow	2a. The User's connection is lost: the message is not received live; UC-10 begins, and after reconnection the message appears in the room history.
Postconditions	The message is visible on the User's screen.

UC-07 View Online Users

Field	Description
Use Case ID	UC-07
Use Case Name	View Online Users
Actor	User

Field	Description
Preconditions	The User has joined a room.
Main Flow	(1) The system displays the usernames of all members currently connected to the room. (2) When a member joins, leaves or disconnects, the server sends an updated list. (3) The client replaces the displayed list.
Alternative / Exception Flow	2a. A member's connection drops without a logout: the server removes that member within 60 seconds.
Postconditions	The displayed list matches the server's list of connected members.

UC-08 Leave Chat Room

Field	Description
Use Case ID	UC-08
Use Case Name	Leave Chat Room
Actor	User
Preconditions	The User has joined a room.
Main Flow	(1) The User selects Leave. (2) The client sends a leave request over the socket. (3) The server removes the connection from the room. (4) The other members see a leave notice and an updated online list. (5) The system displays the room list.
Alternative / Exception Flow	1a. The User closes the browser tab instead: the server detects the closed connection and performs steps 3 and 4.
Postconditions	The User is no longer a member of the room and receives no messages from it.

UC-09 Log Out

Field	Description
Use Case ID	UC-09

Field	Description
Use Case Name	Log Out
Actor	User
Preconditions	The User is logged in.
Main Flow	(1) The User selects Logout. (2) The system invalidates the session token. (3) The server closes the socket connection and removes the User from any room. (4) The system displays the Login page.
Alternative / Exception Flow	2a. The server cannot be reached: the client deletes its copy of the token and displays the Login page; the token expires on the server (SEC-REQ-06).
Postconditions	The session token is invalid. The person is a Guest again.

UC-10 Reconnect After Connection Loss

Field	Description
Use Case ID	UC-10
Use Case Name	Reconnect After Connection Loss
Actor	User
Preconditions	The User is logged in and the socket connection has been lost.
Main Flow	(1) The client detects the loss and shows the state "Reconnecting". (2) The client disables the send control. (3) The client attempts to reconnect with the session token. (4) The server validates the token and accepts the connection. (5) The client rejoins the room the User was in and reloads the recent messages. (6) The client shows the state "Connected" and enables the send control.
Alternative / Exception Flow	3a. Five attempts fail: the client shows "Disconnected" and "Connection lost. Please reload the page." 4a. The token has expired: the server refuses the connection and the client displays the Login page.

Field	Description
Postconditions	The User is connected and a member of the same room, or has been told that the connection could not be restored.

7. Security

The system has three security objectives and eight security requirements. The course requires at least two of each. Numerical values are proposed targets (assumption A-05).

7.1 Security Objectives

ID	Objective	Property
SEC-OBJ-01	Only authenticated users shall be able to access chat rooms, send messages or read messages, and user passwords shall remain confidential even if the database is disclosed.	Confidentiality, authentication
SEC-OBJ-02	Every message shall be attributed to the authenticated user who sent it, and shall be delivered only to members of the room it was sent to.	Integrity, authorization
SEC-OBJ-03	Input supplied by one user shall not be able to execute code in, or disrupt the session of, another user.	Integrity, availability

7.2 Security Requirements

ID	Requirement	Objective	Priority
SEC-REQ-01	The system shall store each password only as a salted bcrypt hash with a cost factor of at least 10, and shall never store, log or transmit back to a client a password in plain text.	SEC-OBJ-01	Must

ID	Requirement	Objective	Priority
SEC-REQ-02	The server shall refuse every socket connection request that does not carry a valid, unexpired session token, and shall process no chat event from an unauthenticated connection.	SEC-OBJ-01	Must
SEC-REQ-03	The server shall set the sender of every message from the identity bound to the authenticated connection, and shall ignore any sender name supplied in a client payload.	SEC-OBJ-02	Must
SEC-REQ-04	The server shall deliver a message only to connections that are members of the target room, and shall reject a send request from a connection that has not joined a room.	SEC-OBJ-02	Must
SEC-REQ-05	The server shall validate the type and length of every client-supplied field (username, password, room name, message text), and the client shall render all user-supplied text as plain text so that HTML or script content is displayed and not executed.	SEC-OBJ-03	Must
SEC-REQ-06	A session token shall be a random value of at least 128 bits, shall expire 24 hours after it is issued, and shall be invalidated immediately on logout.	SEC-OBJ-01	Must
SEC-REQ-07	After 5 consecutive failed login attempts for one username, the system shall reject further login attempts for that username for 5 minutes.	SEC-OBJ-01	Should
SEC-REQ-08	When the system is deployed on any network other than localhost, all client-server traffic shall use TLS (HTTPS and WSS).	SEC-OBJ-01, SEC-OBJ-02	Should

7.3 Security Topics Covered

Topic	Covered by
Authentication	FR-03, SEC-REQ-02, SEC-REQ-07
Authorization	SEC-REQ-03, SEC-REQ-04
Secure communication	SEC-REQ-08
Password protection	SEC-REQ-01, FR-20
Protection against unauthorized access	SEC-REQ-02, SEC-REQ-04, SEC-REQ-07
Input validation	FR-19, SEC-REQ-05
Session and connection security	FR-04, SEC-REQ-02, SEC-REQ-06

7.4 Known Limitations

- Messages are not end-to-end encrypted. The server and anyone with database access can read them.
- On localhost the demonstration runs without TLS. SEC-REQ-08 applies only to deployment on a shared network.
- There is no email verification or password recovery.

8. Requirements Traceability Matrix

Every requirement maps to at least one use case and one test case. The test case IDs are **proposed** by the SRS author so that the Test Plan team has a starting point: one test case per requirement, named after it. The Test Plan team may merge or split them, and should update this matrix when they do.

8.1 Functional Requirements

Requirement ID	Requirement description	Related use case	Related test case
FR-01	Register with username and password	UC-01	TC-FR-01
FR-02	Reject duplicate username	UC-01	TC-FR-02
FR-03	Log in and receive session token	UC-02	TC-FR-03
FR-04	Log out and invalidate session	UC-09	TC-FR-04

Requirement ID	Requirement description	Related use case	Related test case
FR-05	Create a uniquely named room	UC-03	TC-FR-05
FR-06	Display the room list	UC-04	TC-FR-06
FR-07	Join one room	UC-04	TC-FR-07
FR-08	Leave the current room	UC-08	TC-FR-08
FR-09	Send a text message of 1 to 500 characters	UC-05	TC-FR-09
FR-10	Deliver message to all room members without reload	UC-06	TC-FR-10
FR-11	Multiple users in one room; sender name shown	UC-06	TC-FR-11
FR-12	Server-assigned timestamp shown as HH:MM	UC-06	TC-FR-12
FR-13	Store messages; show 50 most recent on join	UC-04	TC-FR-13
FR-14	Online user list, updated on change	UC-07	TC-FR-14
FR-15	Join and leave notices	UC-04, UC-08	TC-FR-15
FR-16	Connection state indicator; send disabled when not connected	UC-10	TC-FR-16
FR-17	Automatic reconnection and room rejoin	UC-10	TC-FR-17
FR-18	Remove disconnected user from online list within 60 s	UC-07	TC-FR-18
FR-19	Reject empty, whitespace-only or over-length message	UC-05	TC-FR-19
FR-20	Generic login failure message	UC-02	TC-FR-20
FR-21	Safe error message on server error	UC-05	TC-FR-21
FR-22	Typing indicator (optional)	UC-05	TC-FR-22
FR-23	Private one-to-one message (optional)	UC-05	TC-FR-23

8.2 Non-Functional Requirements

Requirement ID	Requirement description	Related use case	Related test case
NFR-01	95% of messages shown within 1 s with 20 users	UC-05, UC-06	TC-NFR-01
NFR-02	Login completes within 2 s	UC-02	TC-NFR-02
NFR-03	100% delivery, in order, no duplicates	UC-05, UC-06	TC-NFR-03
NFR-04	Reconnect within 10 s of network restoration	UC-10	TC-NFR-04
NFR-05	2-hour run with 20 users without restart	UC-04, UC-05, UC-06	TC-NFR-05
NFR-06	4 of 5 new users complete first message in 3 min	UC-01, UC-02, UC-04, UC-05	TC-NFR-06
NFR-07	20 concurrent users across 5 rooms	UC-04, UC-05, UC-06	TC-NFR-07
NFR-08	Modular code; install from README in 15 min	All (verified by review, not by a use case)	TC-NFR-08
NFR-09	All Must-priority security requirements met	UC-01, UC-02, UC-05, UC-09	TC-SEC-01 to TC-SEC-06
NFR-10	Works on Chrome, Firefox and Edge	UC-01 to UC-10	TC-NFR-10

8.3 Security Requirements

Requirement ID	Requirement description	Related use case	Related test case
SEC-REQ-01	Passwords stored only as salted bcrypt hashes	UC-01, UC-02	TC-SEC-01
SEC-REQ-02	Socket connection refused without valid token	UC-02, UC-10	TC-SEC-02
SEC-REQ-03	Sender identity taken from the connection, not the payload	UC-05	TC-SEC-03

Requirement ID	Requirement description	Related use case	Related test case
SEC-REQ-04	Messages delivered only to room members	UC-05, UC-06	TC-SEC-04
SEC-REQ-05	Input validated; user text rendered as plain text	UC-01, UC-03, UC-05	TC-SEC-05
SEC-REQ-06	Token is random, expires in 24 h, invalid after logout	UC-02, UC-09	TC-SEC-06
SEC-REQ-07	Lockout for 5 min after 5 failed logins	UC-02	TC-SEC-07
SEC-REQ-08	TLS when deployed beyond localhost	UC-02 to UC-10	TC-SEC-08

8.4 Coverage Summary

Check	Result
Requirements with no use case	None
Requirements with no test case	None
Use cases with no requirement	None
Suggested minimum set for the Test Plan (7 to 10 test cases required by the course)	TC-FR-01, TC-FR-03, TC-FR-07, TC-FR-09, TC-FR-10, TC-FR-17, TC-FR-19, TC-NFR-01, TC-SEC-01, TC-SEC-02

9. Requirements Quality Check

All 41 requirements (23 FR, 10 NFR, 8 SEC-REQ) were reviewed against the eight quality criteria. All pass; Section 9.2 lists four points that affect how they are tested.

9.1 Check by Criterion

Criterion	How it was checked	Result
Clear	Each requirement names the acting party (system, server or client), the action and the object. Terms are defined in Section 1.3.	Pass

Criterion	How it was checked	Result
Unambiguous	No requirement uses words such as fast, easy, user-friendly, secure, robust, adequate or "etc." without a stated measure. Exact message texts and limits are quoted.	Pass
Concise	Each requirement is one or two sentences and states one capability, with its direct failure response where relevant.	Pass
Consistent	Limits are identical wherever they appear: message 1 to 500 characters (FR-09, FR-19); 5 reconnect attempts (FR-17, UC-10); 60 s removal (FR-18, UC-07); 20 users (NFR-01, NFR-05, NFR-07). The terms Guest, User, room, client and server are used with one meaning.	Pass
Testable	Each requirement has an observable pass or fail outcome and a proposed test case in Section 8. Each NFR states its verification method in Section 4.	Pass, see notes 1 and 2
Measurable	Every quality requirement states a number and a unit, or a countable condition.	Pass, see note 3
Feasible	Every Must and Should requirement can be met with Node.js, Socket.IO, SQLite and bcrypt on one machine. Features that risk the schedule are marked Optional.	Pass, see note 4
Uniquely identifiable	IDs FR-01 to FR-23, NFR-01 to NFR-10, SEC-OBJ-01 to SEC-OBJ-03, SEC-REQ-01 to SEC-REQ-08 and UC-01 to UC-10 are each used once.	Pass

9.2 Notes

1. **FR-21** can be tested only by causing a server error on purpose, for example by stopping the database during a request. The Test Plan should describe how the fault is injected.
2. **NFR-08** is verified partly by code review, which depends on the reviewer. The module list in the requirement keeps the review objective.

3. **All numerical targets are proposed**, not given by the project brief (assumption A-05). They should be confirmed with the faculty guide, and may be adjusted after the first load test.
4. **NFR-01, NFR-05 and NFR-07** need 20 simultaneous clients. This requires a small script that simulates clients; 20 real users are not needed.

9.3 Open Points for the Team

- Confirm the technology stack (assumption A-01) and the database (assumption A-02).
- Confirm that registration and login are in scope (assumption A-03).
- Decide whether the optional requirements FR-22 and FR-23 will be attempted.

Appendix A. Consolidated List of Requirements

The system has 23 functional requirements (21 committed, 2 optional), 10 non-functional requirements, 3 security objectives and 8 security requirements. Full wording is in Sections 3, 4 and 7.

A.1 Functional Requirements

ID	Short title	Group	Priority
FR-01	User registration	Account	Must
FR-02	Duplicate username rejection	Account	Must
FR-03	User login	Account	Must
FR-04	User logout	Account	Must
FR-05	Create chat room	Room	Must
FR-06	List chat rooms	Room	Must
FR-07	Join chat room	Room	Must
FR-08	Leave chat room	Room	Must
FR-09	Send message	Messaging	Must
FR-10	Real-time message delivery	Messaging	Must
FR-11	Multi-user rooms with sender name	Messaging	Must
FR-12	Message timestamp	Messaging	Must
FR-13	Message history on join	Messaging	Should

ID	Short title	Group	Priority
FR-14	Online user list	Presence	Must
FR-15	Join and leave notices	Presence	Should
FR-16	Connection state indicator	Connection	Must
FR-17	Automatic reconnection	Connection	Must
FR-18	Offline detection	Connection	Must
FR-19	Invalid message rejection	Error handling	Must
FR-20	Login failure message	Error handling	Must
FR-21	Server error message	Error handling	Should
FR-22	Typing indicator	Optional feature	Optional
FR-23	Private messaging	Optional feature	Optional

A.2 Non-Functional Requirements

ID	Short title	Category	Proposed target
NFR-01	Message delivery time	Performance	95% within 1 s, 20 users
NFR-02	Login response time	Performance	2 s
NFR-03	Message delivery completeness and order	Reliability	100%, in order, no duplicates
NFR-04	Reconnection time	Reliability	10 s after network restoration
NFR-05	Continuous operation	Availability	2 hours, 20 users, no restart
NFR-06	First-use success	Usability	4 of 5 users within 3 min
NFR-07	Concurrent users	Scalability	20 users, 5 rooms
NFR-08	Modular code and documented setup	Maintainability	4 modules; setup in 15 min

ID	Short title	Category	Proposed target
NFR-09	Security compliance	Security	All Must-priority SEC-REQ met
NFR-10	Browser support	Compatibility	Chrome, Firefox, Edge

A.3 Security Objectives and Requirements

ID	Short title	Priority
SEC-OBJ-01	Authenticated access and password confidentiality	Objective
SEC-OBJ-02	Message attribution and room-restricted delivery	Objective
SEC-OBJ-03	Protection against malicious input	Objective
SEC-REQ-01	Password hashing with bcrypt	Must
SEC-REQ-02	Authenticated socket connections only	Must
SEC-REQ-03	Server-assigned sender identity	Must
SEC-REQ-04	Room-restricted message delivery	Must
SEC-REQ-05	Input validation and plain-text rendering	Must
SEC-REQ-06	Session token strength, expiry and invalidation	Must
SEC-REQ-07	Login attempt lockout	Should
SEC-REQ-08	TLS beyond localhost	Should

Appendix B. Assignment Compliance Checklist

The SRS meets all four SRS items in the course handout and all nine sections of the SRS brief. Three items need action by the team before submission; they are marked "Met, with note".

B.1 Course Handout (Mini-Project Deliverables Part-1, item 1: SRS)

Handout requirement	Status	Where
1.a Standard IEEE format	Met, with note 1	Whole document; structure follows IEEE Std 830-1998
1.a.i Introduction and other descriptive sections	Met	Sections 1 and 2
1.a.ii Requirements section with both FRs and NFRs, specified as clear, unambiguous, concise and measurable	Met	Sections 3 and 4; checked in Section 9
1.a.iii UML use case diagram (at least 1)	Met, with note 2	Section 6.4
1.a.iv Security section with at least 2 objectives and 2 requirements	Met (3 objectives, 8 requirements)	Section 7

B.2 SRS Brief

Required item	Status	Where
1. Introduction: purpose, scope, definitions, references, overview	Met	Sections 1.1 to 1.5
2. Overall description: perspective, functions, user classes, environment, constraints, assumptions	Met	Sections 2.1 to 2.6
3. Numbered functional requirements covering registration, login, authentication, create and join chat, send, real-time receive, multiple users, timestamps, online status, connection handling, error handling	Met	Section 3 (FR-01 to FR-23)
Each requirement has a unique ID and uses "shall"	Met	Sections 3, 4, 7

Required item	Status	Where
4. Measurable NFRs for performance, reliability, availability, usability, scalability, maintainability, security, compatibility	Met	Section 4 (NFR-01 to NFR-10)
Numerical values identified as proposed targets	Met	Assumption A-05; Sections 3, 4, 7
5. External interfaces: user, software, communication, hardware	Met	Sections 5.1 to 5.4
Explanation of client-server socket communication	Met	Section 5.3
6. Use case diagram with actors, use cases and relationships	Met, with note 2	Sections 6.1 to 6.4
Use case descriptions with ID, name, actor, preconditions, main flow, alternative flow, postconditions	Met (all 10 use cases)	Section 6.5
7. Security: at least 2 objectives and 2 requirements with IDs	Met	Section 7
8. Traceability matrix: requirement ID, description, use case, test case	Met, with note 3	Section 8
9. Requirements quality check	Met	Section 9
Optional features clearly marked	Met	Section 3.7 (FR-22, FR-23)
Assumptions separated from confirmed features	Met	Section 2.6
Consistent terminology; sections and requirements numbered	Met	Section 1.3; checked in Section 9.1

B.3 Notes

1. **Front matter.** Add a title page with the course name, team members, roll numbers, faculty guide and a revision history table, in the layout your department's IEEE template uses.
2. **Diagram notation.** The Mermaid diagram uses circles for actors because Mermaid cannot draw UML stick figures. If the faculty requires strict UML notation, redraw the same diagram in draw.io, StarUML or PlantUML using Sections 6.1 to 6.3.

3. **Test case IDs.** The test case IDs in Section 8 are proposals. Replace them with the final IDs once the Test Plan is written.